

Clinical Decision Making and Pattern Recognition in Health Care

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June 27, 2026

Introduction and Topic Concept

Healthcare organizations process millions of insurance claims every year, making efficient claim review essential for controlling costs and ensuring appropriate care. Organizations like Cotiviti use advanced payment integrity solutions to help payers identify improper payments and improve claim accuracy. As healthcare claims continue to grow in volume and complexity, AI presents an opportunity to further enhance review efficiency and decision support (Cotiviti, 2024).

Recent advances in AI, including Retrieval-Augmented Generation (RAG), machine learning, anomaly detection, and Agentic AI, are improving healthcare claim review by supporting clinical assessment, billing analysis, fraud detection, and more consistent payment decisions while assisting human reviewers (Salesforce, 2024).

Current Trends in Healthcare AI

Healthcare AI is evolving toward hybrid systems that combine traditional machine learning with Large Language Models. Retrieval-Augmented Generation (RAG) improves LLM responses by retrieving relevant clinical knowledge before generating recommendations, reducing hallucinations and improving explainability (Lewis et al., 2020).

Another emerging trend is Agentic AI, where multiple specialized agents independently perform tasks such as clinical review, financial prediction, provider behavior analysis, and fraud detection before a decision agent combines their outputs. Traditional machine learning remains important for prediction and anomaly detection, particularly when labelled healthcare data is limited (Breiman, 2001; Liu et al., 2008).

Opportunities for Cotiviti

As a proof of concept, I developed an **Agentic Healthcare Payment Integrity Review Assistant** that combines RAG, machine learning, anomaly detection, and LLM reasoning to support healthcare claim review.

The architecture includes four specialized tools for clinical assessment, financial prediction, provider behavior analysis, and fraud detection. Their outputs are combined by a Decision Agent that recommends whether a claim should be approved, manually reviewed, or denied while generating an explainable summary for payment integrity reviewers.

Beyond this proof of concept, similar systems could support automated policy interpretation, provider benchmarking, real-time claim prioritization, and continuous learning from reviewer feedback.

Threats and Challenges

Although AI offers significant opportunities, challenges remain. Large Language Models require reliable knowledge sources to avoid unsupported explanations, making RAG and human oversight essential (Lewis et al., 2020).

Machine learning models may inherit bias from historical claims data, while fraud detection remains difficult because confirmed fraud examples are limited. In addition, healthcare systems must comply with HIPAA requirements when handling sensitive patient information. AI should therefore function as a decision-support tool that assists reviewers while keeping final payment decisions under human oversight (Cognizant, 2024).

Strategic Recommendation

This proof of concept demonstrates how RAG-based clinical reasoning, financial prediction, provider behavior analysis, and anomaly detection can work together within an agentic architecture to support healthcare payment integrity.

Future work could incorporate payer-specific policies, real-time claims processing, and continuous learning from reviewer feedback to improve scalability, consistency, and decision quality across healthcare payment integrity workflows.

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